WHICH IS THE

UNIT RATES / REAL-WORLD SHOPPING

BETTER DEAL?

Compare the prices. Check the quantities.
Make a decision you can explain.

12 COMPARISONS	2 LEVELS	1 BUDGET CHALLENGE

Grades 6-7 | Printable PDF

Whole-dollar and decimal unit rates, realistic shopping choices, a reusable organizer, and a three-question exit ticket.

Includes worked solutions and teacher guidance.

All prices and stores are fictional. No sales tax or extra fees apply.

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Teacher guide

A practical review of unit rates and evidence-based decisions.

Choose a route

Use Level 1 (problems 1-6) for whole-dollar unit rates or Level 2 (7-12) for decimal rates. These are different problems, not duplicate versions. Assign both for a progression.

Suggested pacing

Allow approximately 35-50 minutes for the example, one level, and discussion. Use the budget challenge in a second session or as an extension. Timing will vary; this resource has not been classroom-timed.

Prepare

Print student pages 3-10 as needed. Page 4 is a reusable organizer; pages 5-8 already contain work space. Students need a pencil. Calculators are optional after students set up each division.

Teach the distinction

Lower unit price means better value per unit. The best purchase also depends on quantity needed and budget. Problems 3 and 11 are ties; problem 12 separates unit value from an affordable purchase.

Discuss and assess

Ask students to label units, compare equal quantities, and justify choices. In problem 10 compare per 100 g; in problem 11 compare per 100 mL. The budget challenge requires whole packs.

Page map

3: worked example | 4: organizer | 5-8: practice | 9: budget challenge | 10: exit ticket | 11-14: teacher keys | 15: terms.

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Start with one unit

WORKED EXAMPLE

Name: _____ Date: _____

Store A sells 4 pens for $3.20. Store B sells 6 identical pens for $4.50. Which offer has the lower price per pen?

OFFER	CALCULATION	PRICE PER PEN
A	$3.20 ÷ 4	$0.80
B	$4.50 ÷ 6	$0.75

Decision: Offer B costs $0.05 less per pen.

Why divide? Dividing total price by the number of pens gives the cost of one pen. Now the two offers use the same unit.

Check the purchase: If you need 4 pens and have only $4.00, choose A. Offer B has the lower unit price, but its $4.50 pack exceeds your budget.

Before you decide
1. Compare the same item and quality.
2. Use the same unit for both offers.
3. Divide price by quantity and label the result.
4. Check the budget and required quantity.

All items in each comparison are the same quality. Buy whole packs only. Prices exclude tax and fees. The practice unit rates are exact to the cent; no rounding is needed.

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My comparison organizer

REUSABLE SUPPORT PAGE

Name: _____ Date: _____

Item and comparison unit:

Offer A: quantity, total price, and price ÷ quantity

Offer B: quantity, total price, and price ÷ quantity

Compare the unit prices. Is one lower, or is there a tie?

Check: How many do I need? What is my budget?

My decision and mathematical evidence:

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Level 1: compare the offers

For each problem, calculate both unit rates and explain your choice.

Name: _____ Date: _____

01 Sketchbooks

A: 3 books for $12.00 | B: 5 books for $15.00

Compare dollars per book.

A unit rate: _____ B unit rate: _____

Work and explanation:

02 Glue sticks

A: 4 sticks for $8.00 | B: 6 sticks for $18.00

Compare dollars per stick.

A unit rate: _____ B unit rate: _____

Work and explanation:

03 Ribbon

A: 3 meters for $15.00 | B: 5 meters for $25.00

Compare dollars per meter.

A unit rate: _____ B unit rate: _____

Work and explanation:

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Level 1: compare the offers

For each problem, calculate both unit rates and explain your choice.

Name: _____ Date: _____

04 Markers

A: 6 markers for $12.00 | B: 10 markers for $10.00

Compare dollars per marker.

A unit rate: _____ B unit rate: _____

Work and explanation:

05 Storage boxes

A: 2 boxes for $14.00 | B: 4 boxes for $24.00

Compare dollars per box.

A unit rate: _____ B unit rate: _____

Work and explanation:

06 Notebooks

A: 5 notebooks for $20.00 | B: 3 notebooks for $15.00

Compare dollars per notebook.

A unit rate: _____ B unit rate: _____

Work and explanation:

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Level 2: compare the offers

For each problem, calculate both unit rates and explain your choice.

Name: _____ Date: _____

07 Granola bars

A: 6 bars for $3.90 | B: 10 bars for $6.00

Compare dollars per bar.

A unit rate: _____ B unit rate: _____

Work and explanation:

08 Juice boxes

A: 8 boxes for $4.40 | B: 12 boxes for $7.20

Compare dollars per box.

A unit rate: _____ B unit rate: _____

Work and explanation:

09 Rice

A: 2 kg for $3.80 | B: 5 kg for $9.00

Compare dollars per kg.

A unit rate: _____ B unit rate: _____

Work and explanation:

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Level 2: compare the offers

For each problem, calculate both unit rates and explain your choice.

Name: _____ Date: _____

10 Trail mix

A: 300 g for $2.70 | B: 500 g for $4.25

Compare dollars per 100 g.

A unit rate: _____ B unit rate: _____

Work and explanation:

11 Soap refill

A: 500 mL for $3.75 | B: 900 mL for $6.75

Compare dollars per 100 mL.

A unit rate: _____ B unit rate: _____

Work and explanation:

12 Party cups

A: 8 cups for $2.40 | B: 30 cups for $8.40

Compare dollars per cup. Then choose packs for 24 cups with a $7.50 budget.

A unit rate: _____ B unit rate: _____

Work and explanation:

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The club supply challenge

BUY ENOUGH. STAY WITHIN $20.00.

Name: _____ Date: _____

Your club needs at least 24 snack bars, 24 juice boxes, and 24 pencils. Items are identical within each category. Buy whole packs, mix offers if useful, and do not pay tax or fees. Minimize the total amount spent.

ITEM	OFFER A	OFFER B
Snack bars	6 for $1.80	12 for $3.00
Juice boxes	8 for $2.40	12 for $4.20
Pencils	12 for $3.60	24 for $6.00

1. Calculate all six unit prices. Show your work.

YOUR PLAN	PACKS / TOTAL ITEMS	COST
Snack bars		
Juice boxes		
Pencils		

2. Total spent: _____ Money left: _____

3. A third snack offer has 30 bars for $6.60. It has the lowest price per bar. Should it replace your snack choice if your goal is to spend as little as possible? Support your answer with costs.

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Exit ticket

UNIT RATES AND SHOPPING DECISIONS

Name: _____ Date: _____

1. Offer A: 4 erasers for $2.80. Offer B: 6 erasers for $3.90. Find both unit prices and choose the lower one.

2. One bag holds 250 g of oats for $1.50; another holds 400 g for $2.20. Compare the price per 100 g. Show how you made the units match.

3. You need 10 folders and have $6.00. Offer A: 5 folders for $2.75. Offer B: 12 folders for $6.00. Which has the lower unit price? Which purchase meets your need for the least total cost? Explain.

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Answer key: Level 1

TEACHER COPY / EXACT UNIT RATES

01 Sketchbooks | B

A: $12.00 ÷ 3 = $4.00 per book. B: $15.00 ÷ 5 = $3.00 per book. B has the lower unit price, saving $1.00 per book.

02 Glue sticks | A

A: $8.00 ÷ 4 = $2.00 per stick. B: $18.00 ÷ 6 = $3.00 per stick. A has the lower unit price, saving $1.00 per stick.

03 Ribbon | Tie

A: $15.00 ÷ 3 = $5.00 per meter. B: $25.00 ÷ 5 = $5.00 per meter. Both offers have the same unit price.

04 Markers | B

A: $12.00 ÷ 6 = $2.00 per marker. B: $10.00 ÷ 10 = $1.00 per marker. B has the lower unit price, saving $1.00 per marker.

05 Storage boxes | B

A: $14.00 ÷ 2 = $7.00 per box. B: $24.00 ÷ 4 = $6.00 per box. B has the lower unit price, saving $1.00 per box.

06 Notebooks | A

A: $20.00 ÷ 5 = $4.00 per notebook. B: $15.00 ÷ 3 = $5.00 per notebook. A has the lower unit price, saving $1.00 per notebook.

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Answer key: Level 2

TEACHER COPY / EXACT UNIT RATES

07 Granola bars | B

A: $3.90 ÷ 6 = $0.65 per bar. B: $6.00 ÷ 10 = $0.60 per bar. B has the lower unit price, saving $0.05 per bar.

08 Juice boxes | A

A: $4.40 ÷ 8 = $0.55 per box. B: $7.20 ÷ 12 = $0.60 per box. A has the lower unit price, saving $0.05 per box.

09 Rice | B

A: $3.80 ÷ 2 = $1.90 per kg. B: $9.00 ÷ 5 = $1.80 per kg. B has the lower unit price, saving $0.10 per kg.

10 Trail mix | B

300 g and 500 g contain 3 and 5 units of 100 g. A: $2.70 ÷ 3 = $0.90 per 100 g. B: $4.25 ÷ 5 = $0.85 per 100 g. B has the lower unit price, saving $0.05 per 100 g.

11 Soap refill | Tie

500 mL and 900 mL contain 5 and 9 units of 100 mL. A: $3.75 ÷ 5 = $0.75 per 100 mL. B: $6.75 ÷ 9 = $0.75 per 100 mL. Both offers have the same unit price.

12 Party cups | B

A: $2.40 ÷ 8 = $0.30 per cup. B: $8.40 ÷ 30 = $0.28 per cup. B has the lower unit price, saving $0.02 per cup. Buy 3 A packs: 24 cups for $7.20, leaving $0.30. B costs $8.40 and exceeds the $7.50 budget.

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Answer key: club challenge

TEACHER COPY / PURCHASE PLAN

ITEM	A UNIT PRICE	B UNIT PRICE
Snack bars	$0.30 per bar	$0.25 per bar
Juice boxes	$0.30 per box	$0.35 per box
Pencils	$0.30 per pencil	$0.25 per pencil

MINIMUM-COST PLAN	QUANTITY	COST
2 B snack packs	24 bars	$6.00
3 A juice packs	24 boxes	$7.20
1 B pencil pack	24 pencils	$6.00
TOTAL / MONEY LEFT	24 of each	$19.20 / $0.80

Why this is the minimum: Each chosen offer has the lowest unit rate in its category and provides exactly 24 items. Any plan supplying at least 24 must cost at least 24 times that category's lowest unit rate. This plan reaches those lower bounds.

The jumbo snack offer: $6.60 ÷ 30 = $0.22 per bar. This is lower than $0.25 per bar, but costs $0.60 more in total than two B packs. The revised plan costs $19.80 and leaves $0.20, with 6 extra bars. It fits the budget but does not minimize spending. Keep the original plan for the stated goal.

Discussion: If the club valued extra snacks for a later meeting, the jumbo pack could be reasonable. That is a different goal from minimizing today's spending.

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Exit ticket key & feedback

TEACHER COPY

1. Erasers

A: $2.80 ÷ 4 = $0.70 per eraser. B: $3.90 ÷ 6 = $0.65 per eraser. Choose B for the lower unit price.

2. Oats

250 g contains 2.5 units of 100 g: $1.50 ÷ 2.5 = $0.60 per 100 g. For 400 g: $2.20 ÷ 4 = $0.55 per 100 g. The 400 g bag has the lower unit price. Alternatively calculate price per gram, then multiply by 100.

3. Folders

A: $2.75 ÷ 5 = $0.55 per folder. B: $6.00 ÷ 12 = $0.50 per folder. B has the lower unit price. Buy two A packs for $5.50 to get exactly 10 folders and spend the least. B also fits the budget, but costs $0.50 more.

Suggested scoring: 6 points

Q1: 1 point for both rates, 1 for B. Q2: 1 for both rates in the same unit, 1 for the 400 g bag. Q3: 1 for both rates and B as the lower rate, 1 for two A packs at $5.50 with reasoning. Accept equivalent valid methods.

Feedback prompts

Quantity ÷ price? Ask: Does your result mean dollars per item or items per dollar? Comparing sticker prices only? Ask: How much would one item cost? Ignoring budget? Ask: What will you actually pay at checkout?

Budget challenge: suggested 4-point rubric

1: all six unit rates correct. 1: a plan providing at least 24 of each within $20. 1: correct total and change. 1: explains why the jumbo pack has a lower unit rate but a higher total cost.

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